COURSE TITLE : **BUSINESS MATHEMATICS**
COURSE CODE : **2252**
SEMESTER : **2**
COURSE CATEGORY : **F**
PERIODS/WEEK : **5**
PERIODS/SEMESTER : **75**
CREDITS : **5**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	**MATRICES**	15
2	**PROBABILITY**	20
3	**DIFFERENTIATION**	20
4	**INTEGRATION**	20

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To understand the concept of matrices
	2	To understand the applications of matrices
2	1	To understand concept of probability
	2	To understand the uses and applications of probability
3	1	To understand the concept and uses of differentiation
	2	To understand the theorems in differentiation
4	**1**	To understand concept and application of integrations

SPECIFIC OUTCOMES

1.1.0 Understand the concept of matrices
1.1.1 Explain the idea of determinant of a square matrix.
1.1.2 State the properties of determinant.
1.1.3 Solve practical problems by applying these properties like:
1.1.4 Differentiate different types of matrices
1.1.5 Define different operations of matrices. 1. Equality, 2. Addition of matrices, 3. Subtraction, 4. Multiplication by a scalar.
1.1.6 Define the multiplication between two matrices. Do problems.
1.1.7 State the condition for conformability for multiplication
1.1.8 Define the transpose of a matrix
1.1.9 State theorems on transpose.
1.1.10 Define symmetric and skew symmetric matrices

1.1.11 Solve practical problems involving transpose, symmetric and skew symmetric matrices
2.1.0 Understand the concept of probability.
2.1.1 Define random experiment.
2.1.2 Define event, sample space.
2.1.3 Define different types of events
2.1.4 Explain Mutually exclusive events
2.1.5 Explain Equally likely events
2.1.6 Explain Exhaustive events
2.1.7 Explain Independent events
2.1.8 Explain Dependant events
3’1.0 Understand the concept differentiation
3.1.1 State differentiation from first principle
3.1.2 Solve problems using first principle
3.1.3 Find the derivative of standard forms
3.1.4 Derive basic theorems of differentiation.
3.1.5 Deduce product rule, quotient rule of differentiation
3.1.6 Solve problems using the above rule
3.1.7 Differentiate using different forms

1) Chain rule (function of function rule)
2) Implicit function
3) Parametric differentiation
4) Logarithmic differentiation

4.1.0 Understand the Concept of Integration
4.1.1 State fundamental results of Integration
4.1.2 Find the integrals of standard forms
4.1.3 Find integrals using different methods

1) Integration of products and quotients
2) Integration by substitution

Integrals of the form

1)f (ax+b) dx 2) f (xn)xn-1 dx

3)[f(x)]n f(x)dx 4) f ((x)) (x) dx

CONTENT DETAILS

MODULE I: Matrices, Different types of matrices, operation on matrices – equality addition multiplication etc., Transpose, Symmetric and skew symmetric matrices, Determinants Properties.

MODULE II: Probability, random experiment, sample space, events, different types of events,

MODULE III: Differentiation from first principle, product rule, quotient rule, function of function rule, implicit functions, logarithmic differentiation.

MODULE IV: Integral, methods of integration, integration through partial fraction,

BOOKS RECOMMENDED

1. N. Balakrishnan, Business Mathematics, Himalaya publication
2. Prof. Jananrdhanan Pillai, Technical Mathematics, Akhil